

- 6 (a) (i) to reduce power loss in the core
due to eddy currents/induced currents B1 [2]
B1
- (ii) *either* no power loss in transformer
or input power = output power B1 [1]
- (b) *either* r.m.s. voltage across load = $9.0 \times (8100 / 300)$ C1
peak voltage across load = $\sqrt{2} \times 243$
= 340 V A1 [2]
(C1)
- or* peak voltage across primary coil = $9.0 \times \sqrt{2}$
peak voltage across load = $12.7 \times (8100/300)$
= 340 V (A1)